PROJECT TOPIC: ECOWARE — SMART GARBAGE COLLECTION & CITIZEN COMPLAINT MANAGEMENT SYSTEM

Group No.: 79

Project Group Members:

1. Harshit Maheshwari (University Roll No.: 2315000937)

2. Ishu Srivastava (University Roll No.: 2315001000)

3. Manvendra Pratap Singh (University Roll No.: 2315001329)

4. Palak Bhardwaj (University Roll No.: 2315001544)

5. Shashank Shekhar (University Roll No.: 2315002042)

Project Supervisor: Ruchi Gupta

About the Project: EcoWare is a web-based garbage alert and monitoring platform that improves communication between citizens and municipal sanitation authorities. Residents can receive locality-based collection alerts and report missed collections, overflowing bins, and illegal dumping through text, images, or voice messages. A complaint tracker lets users follow reports from submission to resolution. The municipal dashboard enables authorized staff to review complaints, update status, assign action, and monitor collection performance. A proposed community-validation feature uses confirmations from registered households to help verify collection completion. Unresolved complaints can be escalated to supervisors after a defined deadline. AI-assisted features can summarize feedback, categorize complaints, and flag possible duplicate reports, while final verification and official action remain with authorized staff. The responsive platform will be developed using full-stack web technologies and a database.

Motivation: Unpredictable collection schedules and slow complaint handling can contribute to roadside dumping and reduce public trust. EcoWare offers a structured way to send alerts, report local issues, validate services, and track resolutions. Its dashboard can help authorities identify recurring problems and monitor response times. The project is suitable as a final-year major project because it combines frontend and backend development, database management, authentication, role-based access, and data visualization, with AI-assisted feedback analysis as a supporting feature.

Innovation:

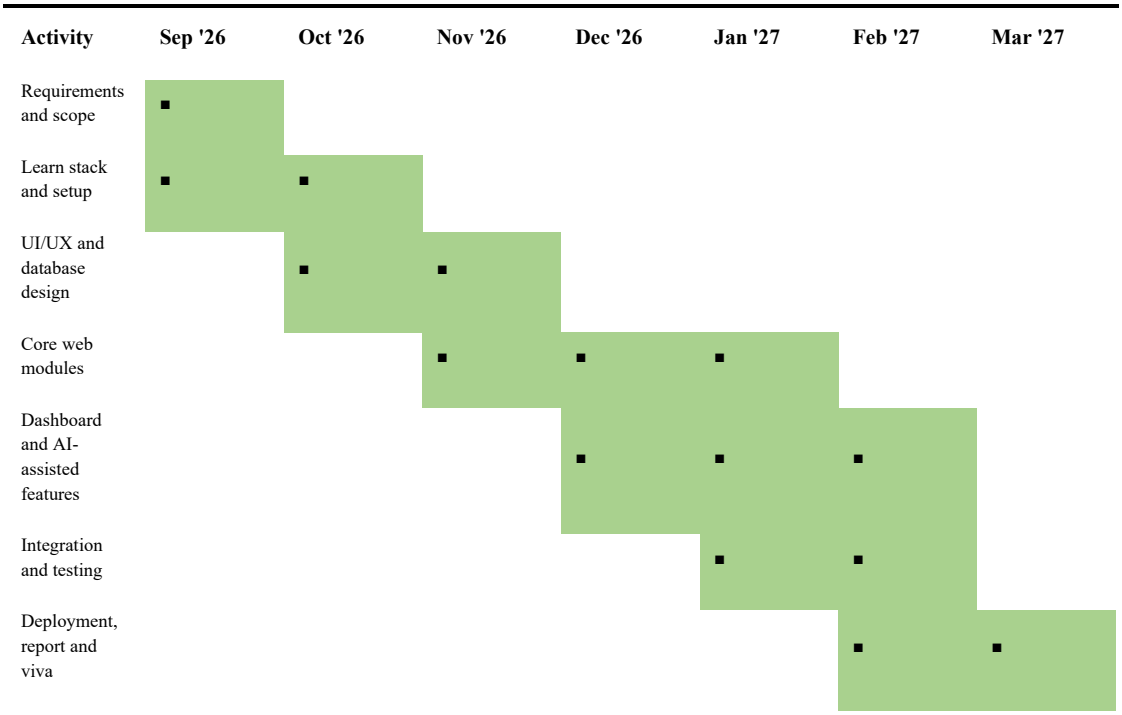
• Locality-based collection alerts help residents plan waste disposal.

• Community validation compares collection status with household confirmations.

• A dashboard highlights recurring complaints, locations, pending issues, and resolution time.

• AI-assisted feedback summaries and duplicate-report detection help staff prioritize complaints.

Project Planning: Work will proceed through requirement analysis, technology learning and setup, UI/UX and database design, core module development, dashboard and AI-assisted feature integration, testing, deployment, and documentation. The proposed schedule below runs from September 2026 to March 2027 for the B.Tech. CSE Session 2026–27.



Tools required:

➢ **Hardware Requirements:** Laptop/desktop (minimum 4 GB RAM recommended), internet connection, and a modern web browser.

➢ **Software Requirements:** Windows/Linux, Visual Studio Code, Node.js, Express.js, MongoDB, Mongoose, EJS, HTML, CSS, JavaScript, Bootstrap, Chart.js, Git/GitHub, and a web browser. An AI service/API may be integrated for complaint categorization and feedback summarization.

Signature of Project Supervisor: _____